

Improving Charge Transport via Intermediate-Controlled Crystal Growth in 2D Perovskite Solar Cells

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Abstract

Reduced-dimensional hybrid perovskite semiconductors have recently attracted significant attention due to their promising stability and optoelectronic properties. However, the issue of poor charge transport in 2D perovskites limits its application. Here, we report our studies on intermediate-controlled crystal growth to improve charge carrier transport in two-dimension (2D) perovskite thin films. We show that the coordination strength of solvents with perovskite precursor affects the initial state of intermediate phase formation as well as the subsequent perovskite layer growth. Tuning the solvent composition with a mixture (5:5) of dimethyl formamide (DMF) and dimethyl sulfoxide (DMSO) leads to the growth of highly-orientated 2D perovskite films with much-improved optoelectronic properties (faster transport by $\sim 50\times$, longer carrier lifetime by $\sim 5\times$, and lower defect density by $\sim 30\times$) than the film prepared with pure DMF. Consequently, perovskite solar cells based on DMF/DMSO (5:5) show $>80\%$ efficiency improvement than the devices based on pure DMF.

Introduction

Organic-inorganic lead halide perovskites are attractive to a variety of optoelectronic applications due to their unique advantages, such as high absorption coefficients, excellent carrier transport, low cost, tunable compositions, and excellent structures, thereby allowing easy fabrication by various processes.^[1-5] In particular, PSCs have achieved a certified PCE of 23.7%, which is unprecedented among all thin-film photovoltaic (PV) technologies.^[6] However, some important obstacles could limit their commercialization.^[7,8] Hole transport materials (HTM) and back-electrode materials used in research laboratories are often too expensive, which need to be replaced by low-cost, abundant and possibly environmentally sustainable alternatives.^[9] Efficiency loss is also inevitable when the solar cell or module area increases, although remarkable progress with large-scale PSCs manufacturing techniques have been demonstrated.^[10] The most important gap is the issue associated with long-term stability. Right now, PSCs would not be ready to work for 25 years in outdoor conditions, since moisture, heat, and light are still challenges hindering PSCs from practical applications.^[11] Researchers have paid attention to experimental and theoretical research activities, aiming at maximizing PSCs performance by choosing materials with suitable elements, ions and functional groups.^[12-14]

Compared to above three-dimensional (3D) perovskites, 2D perovskite crystals have recently attracted an increasing attention due to some unique properties for improving stability.^[15] The hydrophobic bulky alkylammonium cations in 2D perovskite lattices can block the accessible pathways of moisture invasion, making them promising candidates for optoelectronic devices.^[16] However, 2D PSCs generally exhibit large exciton binding energy, low absorption coefficient, and poor charge transport.^[17-19]

The issue of poor charge transport in 2D perovskites is primarily associated with the bulky organic cation spacer between 2D sheets. Strategies to address this issue include controlling perovskite growth with more preferred vertical orientation or enhancing out-of-plane transport with new organic spacer design.^[20,21] For controlling perovskite growth with more preferred vertical orientation, researchers often use hot-casting deposition method,^[22-25] incorporate additives to assist perovskite growth,^[26,27] or change the A-sites to alter the crystal growth orientation.^[28] Although hot-casting has led to high-quality 2D perovskite films, it is difficult to have deterministic and precise control during fabrication.

It is generally preferred to have large grain size, oriented crystal growth, and enhanced charge carrier mobility for the 2D perovskite layer to function in efficient PSCs. Previous studies have emphasized the importance of the intermediate phases in 3D perovskite crystallization,^[29,30] but the effect of controlling the intermediate phases on the growth and charge transport properties of 2D perovskite thin films has not yet been reported.

In this paper, we present studies of charge transport correlated with intermediate-controlled crystal growth in 2D perovskite thin films. The relationship between the morphology, evolution of the perovskite formation, and PV performance are investigated in detail. We show that the coordination strength of solvents with perovskite precursor affects the initial state of the intermediate phase formation as well as the subsequent perovskite layer growth. We demonstrate the growth of highly-oriented 2D perovskite films with much improved optoelectronic properties (faster transport by $\sim 50\times$, longer carrier lifetime by $\sim 5\times$, and lower defect density by $\sim 30\times$) by adjusting the solvent composition. With mixed solvent of DMF/DMSO (5:5), the PSCs show $>80\%$ efficiency improvement than the device prepared with pure DMF.

Results and Discussion

It is well known that the performance of PSCs strongly depend on the film quality and morphology of the perovskite layer.^[31,32] Various mixing ratios of DMF/DMSO ranging from 10:0 to 0:10 were used to study the impact of solvent tuning on the perovskite morphology. Solvents with oxygen donor such as DMF can form adduct with PbI_2 , but more stable adduct is expected from DMSO due to stronger polarity (relative polarity of 0.444 for DMSO vs. 0.386 for DMF).^[33] We conducted scanning electron microscopy (SEM) to study the surface and cross-sectional morphology of the $(\text{PEA})_2\text{MA}_4\text{Pb}_5\text{I}_{16}$ perovskite films as shown in Figures 1 and S1. The thicknesses of these films are about 460 nm, except for the sample prepared with 0:10 DMF/DMSO, which is 380 nm. The perovskite film fabricated from precursor with pure DMF (referred to as 10:0; Figure 1a) shows a homogenous structures and no discernable structures, which is consistent with a high degree of disorder. With increasing the amount of DMSO (DMF/DMSO changing from 10:0 to 5:5), large oriented striations indicative of grains (larger than $1\ \mu\text{m}$) with vertically aligned boundaries were identified in the cross-section SEM

image (Figures 1a and S1). However, further increasing the amount of DMSO resulted in less-ordered grain structures. The transition of the orientation and order with increasing amount of DMSO is more clearly revealed in the 2D X-ray diffraction (XRD) patterns as shown in Figure 1b. When the perovskite film was prepared with DMF/DMSO = 10:0, a relatively broad peak indicative of a highly disordered material is evident, but for DMF/DMSO = 5:5, we observed clear texturing of the main diffraction peaks. Figure 1c shows the schematic illustration of the growth principle with different DMF/DMSO solvents. Similar results could be concluded from the XRD patterns as shown in Figure S2. It shows that strong diffraction signals at $2\theta = 14.1^\circ$ and 28.4° , which correspond to the (111) and (202) lattice planes, respectively.^[34] Compared to the perovskite film fabricated by pure DMF, the films fabricated with added DMSO yield remarkably enhanced diffraction intensities at 14.1° and 28.4° . As the DMSO ratio is increased from 10:0 to 5:5, the full-width at half-maximum (FWHM) of the (111) plane is greatly reduced from 1.00° to 0.32° , as summarized in Table S1. With increasing DMSO ratio to 5:5, the preferential crystal growth of the (111) plane was observed for typical vertical orientation to the substrate surface, which means that perovskite layer could favor perpendicular crystal growth with improved crystallinity.

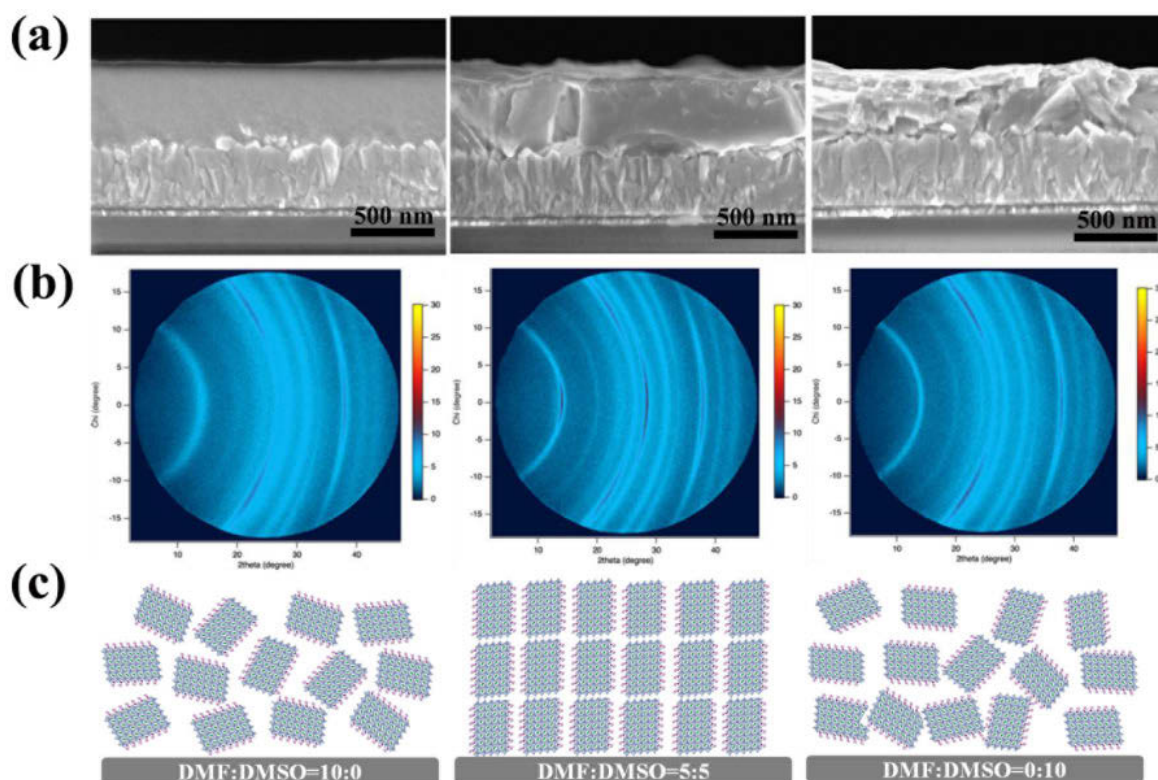


Figure 1. (a) Cross-section SEM images and (b) the corresponding 2D XRD patterns of the $(\text{PEA})_2(\text{MA})_4\text{Pb}_5\text{I}_{16}$ films with different solvent ratio. (c) Schematic representation of $(\text{PEA})_2(\text{MA})_4\text{Pb}_5\text{I}_{16}$ crystal growth with different solvents.

We hypothesize that the reaction kinetics of perovskite layer was related to the interaction strength between PbI_2 and ligand (DMF or DMSO). Both DMF and DMSO can interact with perovskite precursor components to form an intermediate complex during the initial stage of the perovskite formation. This is evidenced by the low-angle ($< 10^\circ$) diffraction peaks (Figure S3). However, DMF has a weak coordination with Pb^{2+} in precursors, which allow a rapid reaction among PbI_2 , PEA, and MAI to form perovskite. This is evident in the time-dependent optical images and UV-vis spectra (Figures S4 and S5), where there is already coexistence of perovskite structure and intermediate complex (e.g., Pb -centered ion-cages $(\text{MAI})_x(\text{PEAI})_y\text{PbI}_2(\text{DMF})_z$) right after antisolvent extraction and before thermal annealing. In the annealing process, the intermediate complex $(\text{MAI})_x(\text{PEAI})_y\text{PbI}_2(\text{DMF})_z$ and present perovskite will lead to different direction-growth processes, which would result in structural mismatch and tend to produce more horizontal boundaries. In addition, fast formation of perovskite crystals could generate a large number of small grains with poor quality without alignment (Figure 2a). No significant changes in the optical properties are observed between the initial and final stage of the resulting perovskite films. In contrast, the strong ligand DMSO constituted with Pb^{2+} to a more stable complex, which requires a longer process to release Pb^{2+} and form the final perovskite structure. This is clearly supported by the optical images and UV-vis absorption spectra for perovskite films prepared with mixed DMF/DMSO (5:5) as shown in Figures S4 and S5. There is only an intermediate complex (e.g., $(\text{MAI})_a(\text{PEAI})_b\text{PbI}_2(\text{DMF})_c(\text{DMSO})_d$) formation right after anti-solvent extraction and before annealing. The intermediate complex would presumably only lead to the up-growth of perovskite and result in a conformal $[110]$ alignment, which would achieve a better interfacial contact and significantly reduced horizontal boundaries. The low nucleation density also leads to large grains at a later crystallization stage (as shown schematically in Figure 2b).^[35] These indeed provide means to further control the crystallization kinetics of 2D perovskite film. The resultant perovskite film made from the optimized ratio of DMF and DMSO (5:5) precursor showed densely packed large grains, where the grains and associated boundaries are vertically

aligned, leading to a higher quality and strongly textured film. When too much DMSO was used, Pb^{2+} has difficulty reacting with cations (MA and PEA), which could account for the reduced film quality of the perovskite layer.^[36,37]

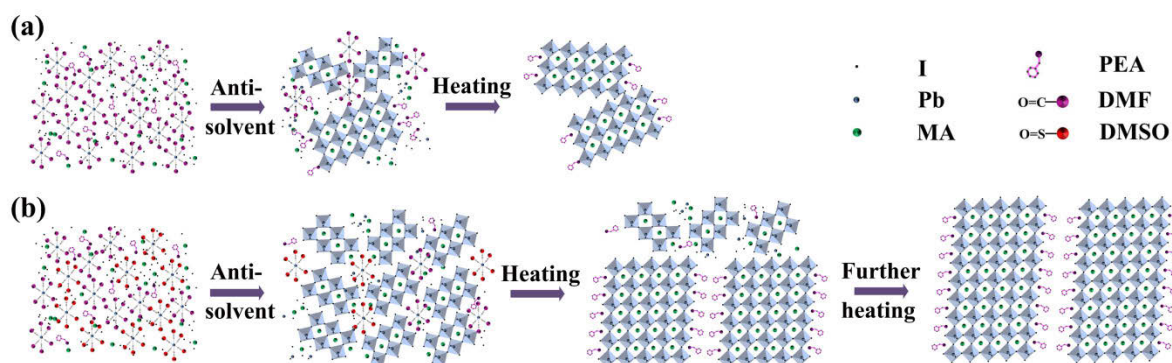


Figure 2. Schematic illustration of 2D perovskite crystal growth in (a) pure DMF solvent and (b) mixed solvent DMF/DMSO (5:5).

To gain further insight into the photophysics of 2D perovskite layers, a variety of spectroscopy techniques (i.e., photoluminescence (PL), and time-resolved photoluminescence (TRPL) and microwave conductivity (TRMC)) were conducted. FTO/2D perovskite/spiro-OMeTAD films fabricated from DMF/DMSO (5:5) showed much lower PL intensity than those fabricated from pure DMF. Note that the spiro-OMeTAD is a standard hole transport material used in perovskite solar cells. Thus, the much-lower PL intensity prepared with DMF/DMSO (5:5) is likely caused by the more effective charge quenching/separation resulting from improved charge transport/collection process in comparison to those prepared with pure DMF. This is consistent with the proposed growth mechanism (Figure 1c), where vertically aligned grains are beneficial for charge transport and collection. TRPL was used to examine the recombination lifetime in perovskite thin films deposited on glass substrates. Consistent with the above results, perovskite films fabricated from DMF/DMSO (5:5) shows longer carrier lifetime (about 100 ns) compared to the lifetime of ~20 ns for the perovskite films fabricated from pure DMF.^[38,39] Most strikingly is the TRMC measurement of the carrier mobility. Figure 3c compares the typical TRMC transients with 640-nm laser excitation for perovskite thin films prepared from pure DMF and DMF/DMSO (5:5). The yield-mobility product $\phi\Sigma\mu$ value is increased from $0.08 \text{ cm}^2 \text{ V}^{-1}\text{s}^{-1}$ for DMF to $4.3 \text{ cm}^2 \text{ V}^{-1}\text{s}^{-1}$ for DMF/DMSO (5:5) though grains below a certain critical diameter constrain the true mobility value.^[40] Figure 3d shows the

comparison of the $\phi\Sigma\mu$ value over a wide range of laser excitation intensity. This dramatic improvement of the transport property is consistent with the much reduced dark carrier density ($n_d = 2.2 \times 10^{17} \text{ cm}^{-3}$) for the DMF/DMSO (5:5) sample compared to the DMF sample ($n_d = 6.2 \times 10^{18} \text{ cm}^{-3}$), as shown in Figure S6. These results indicate that by controlling the reaction between Pb^{2+} and solvent, the quality of perovskite film can be much improved with lower non-radiative recombination, longer carrier lifetime, and faster transport.

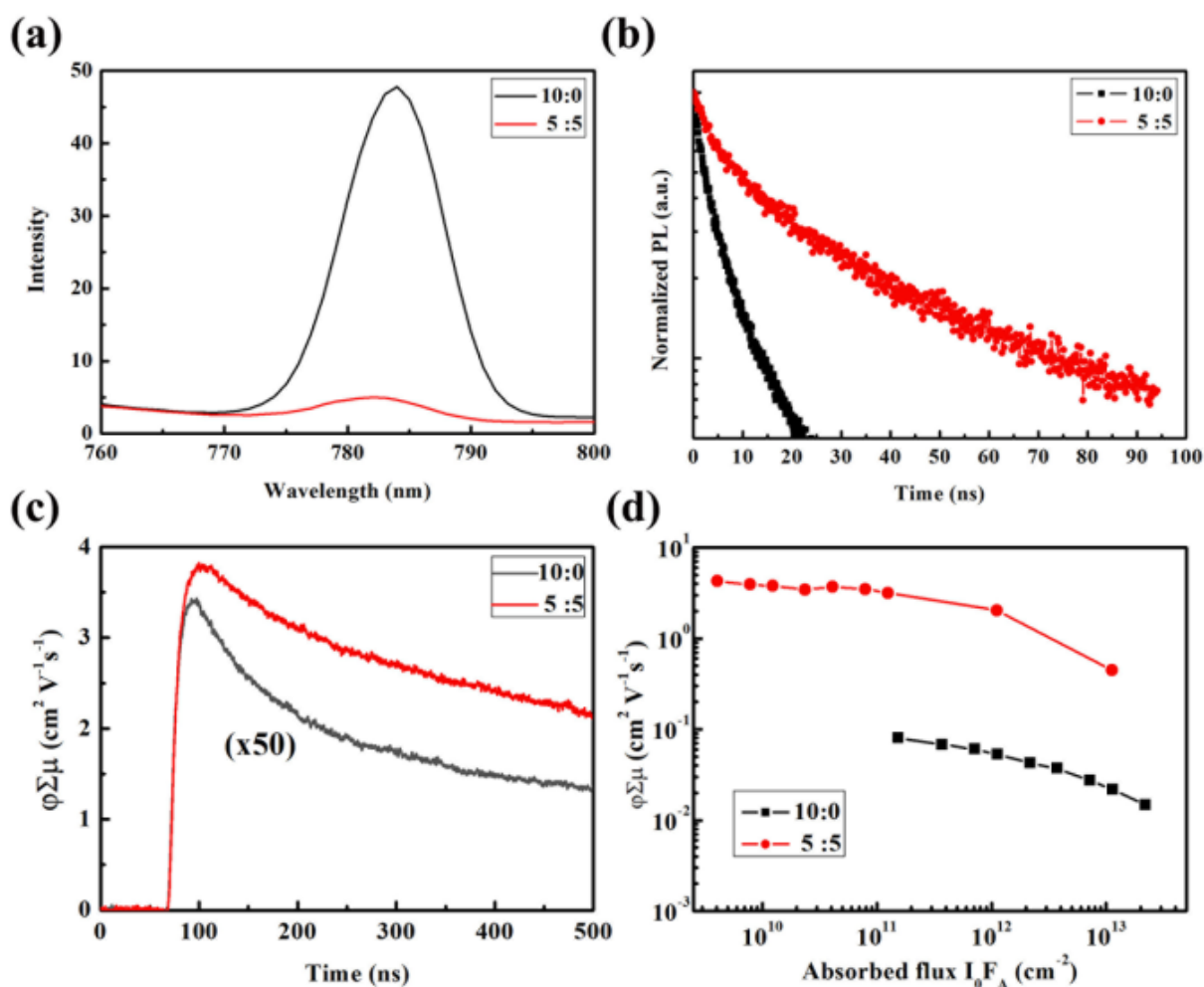


Figure 3. (a) PL spectra of PSCs. (b) TRPL spectra of perovskite films deposited on glass. (c) Typical TRMC transients and (d) intensity dependence of perovskite thin films prepared with 10:0 and 5:5 DMF/DMSO solvents.

We have also carried out conductive atomic force microscopy (C-AFM) to further verify the conducting properties of the corresponding 2D perovskite films, and the results are shown in Figure 4. The 2D perovskite film fabricated by pure DMF showed relatively low current signal, which indicates poor electrical conductivity (less than 30 pA). However, when

DMF/DMSO (5:5) was used, the sample displayed a significantly increased (about 100-fold) current signal, which confirms better carrier conduction. This drastic difference is further illustrated in the line-profile comparison of the C-AFM results (Figure 4c and d).

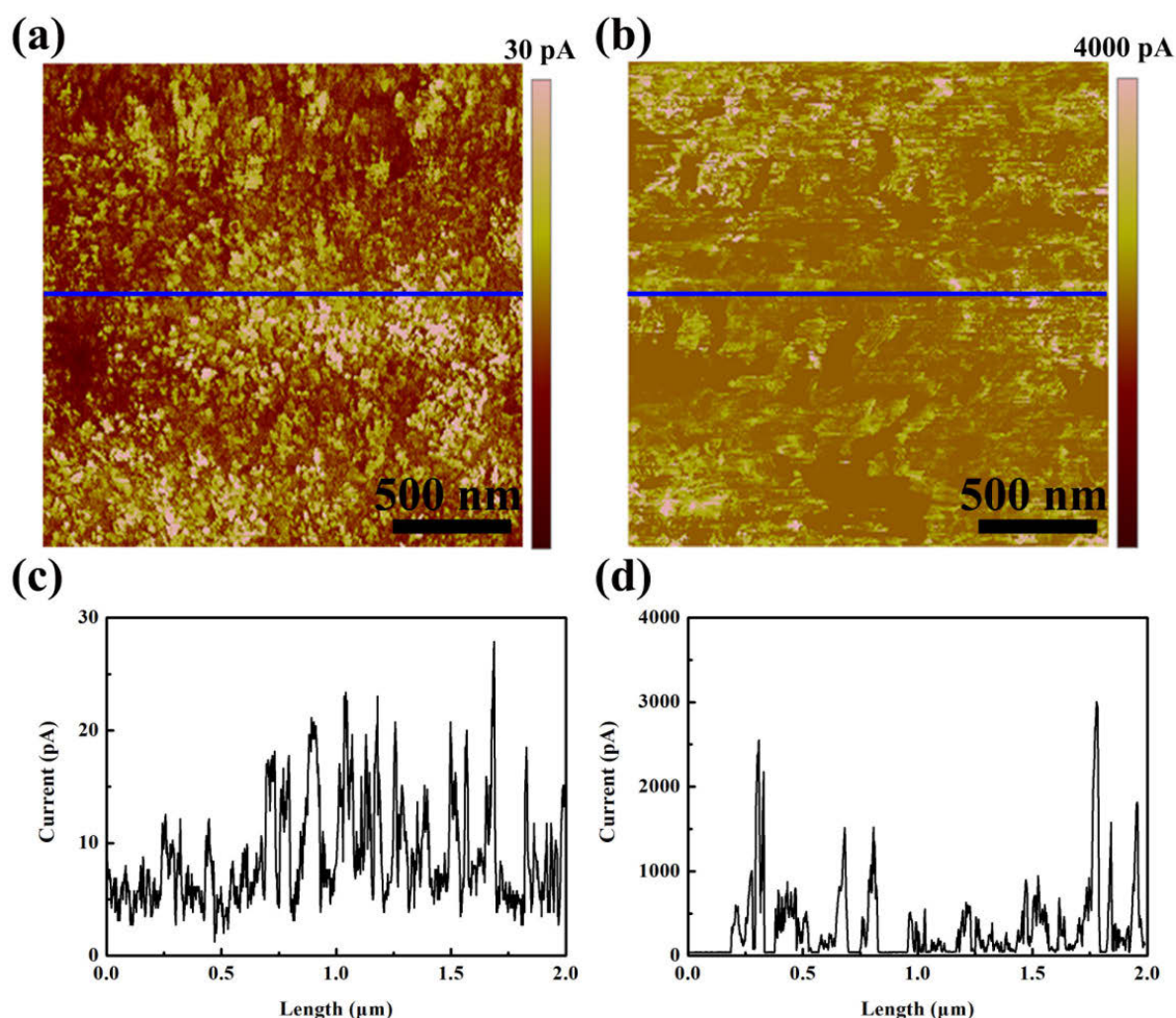


Figure 4. The C-AFM mapping images for the perovskite films at a bias voltage of 2 V in the dark. The solvent ratio of DMF/DMSO is 10:0 in (a) and 5:5 in (b). The corresponding line profile comparison for samples in (a) and (b) are shown in (c) and (d), respectively.

To illustrate the impact of different solvent ratios in 2D perovskite precursor on PV performance (device stack: glass/FTO/compact-TiO₂/perovskite/spiro-MeOTAD/Au), the photocurrent density–voltage (J–V) curves of these PSCs were recorded under standard AM1.5G illumination (100 mW cm⁻²), as shown in Figure S7. The corresponding PV parameters are summarized in Table S2. Compared to the pure DMF solvent precursor, it can be found that when DMSO content increases from (10:0) to (5:5), the PCE of PSCs sharply

increases by >80%, from 6.78% to 12.29% (Figure 5a). All PV parameters including the short-circuit current (J_{sc}), open-circuit voltage (V_{oc}), and fill factor (FF), are all improved by a wide margin, which can be attributed to the improved structural, optical and electrical properties. In particular, the significant enhancement of FF from 45% to 67% supports the benefit of growing more oriented 2D perovskites. This intermediate-controlled crystal growth method could be widely used in 2D PSCs. It is not only suitable to apply in PEA based 2D perovskite material, but also can be utilized in other 2D materials such as butylammonium (BA) based as shown in Figure S8. The incident photon-to-electron conversion efficiency (IPCE) plots for corresponding devices are shown in Figure 5b. Integration of the IPCE spectrum gives a photocurrent density, which is in good agreement with the J_{sc} extracted from the J-V measurement. The stabilized power outputs (SPO) measurement is required when the perovskite solar cell display hysteresis as shown in Figure 5a. This approach can eliminate transient behavior (e.g., capacitive charging/discharging and ion migration) that can affect the determination of the device efficiency. The consistent SPO results (Figure 5c) can further confirm the accuracy of the J-V measurement. One important aspect of perovskite solar cell research and development is manufacturability (production capacity and yields) without large batch-to-batch variations. Figure 5d shows that PSCs devices have good reproducibility in this study. The average PCE of devices, obtained from the measurements of 16 devices, based on mixed solvent DMF/DMSO (5:5) is about 12%. The highest efficiency is 12.29%. In contrast, the average PCE of devices based on pure DMF solvent is about 6.5%.

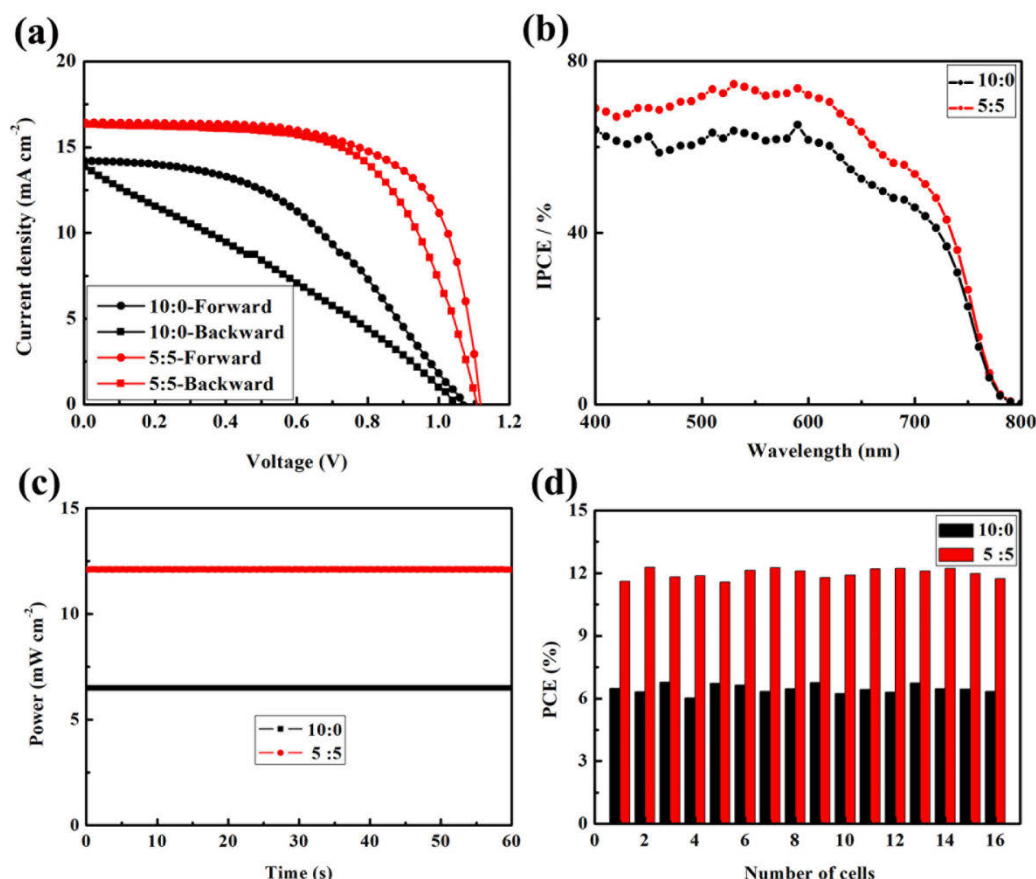


Figure 5. (a) J-V curves, (b) IPCE spectra, (c) stabilized power outputs, and (d) PCE histogram of PSCs based on solvent with different DMF and DMSO ratio as indicated, the red histogram in (d) represents devices based on mixed solvent DMF/DMSO (5:5), and the black histogram in (d) represents devices based on pure DMF solvent.

Conclusions

In summary, we show that charge transport in 2D perovskite films can be significantly improved by intermediate-controlled crystal growth via solvent tuning of the perovskite precursor solution. The formation of the intermediate phase during the initial state of the film formation strongly affects the subsequent perovskite layer growth. Tuning the solvent composition with a mixture (5:5) of dimethyl formamide (DMF) and dimethyl sulfoxide (DMSO) leads to the growth of highly-orientated 2D perovskite films with much faster transport, longer carrier lifetime, and lower defect density than the film prepared with pure DMF. The improvement in these optoelectronic properties lead to >80% efficiency improvement in perovskite solar cells.

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